Jonathan Hsiao

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EDUCATION

The Pennsylvania State University — University Park, PA
B.S. Computer Science, College of Engineering — Expected May 2027
Minors in Artificial Intelligence and Quantum Information Science — **GPA: 3.99 / 4.00**
Dean's List 2024-2026 | President Walker Award and President Sparks Award (sustained 4.00 GPA)

TECHNICAL SKILLS

- **Languages:** Python, C, R, Java, JavaScript, Verilog, MIPS Assembly
- **ML / AI:** PyTorch, TensorFlow, Reinforcement Learning, Deep Learning, CNNs, YOLOv8, Federated Learning, Multi-agent network, LLM APIs, VLA, RAG
- **Software and Tools:** FastAPI, React, ChromaDB, PyQt6, NumPy, Pandas, Git, Linux
- **Relevant Courses:** Machine Learning | Algorithms | Computer Architecture | Systems Programming

RESEARCH EXPERIENCE

Semiconductor Spectroscopy Lab, Penn State — University Park, PA
Research Assistant — Mar 2025 – Present

- Refined the PID signal analysis model on a $300,000 Amazon-funded quantum computing project, cutting convergence time **15%** and raising signal-to-noise ratio **10%**.
- Constructed the PyQt6 acquisition and control software for CW EPR and EDMR spectrometers: a digital lock-in amplifier and an adaptive signal averaging suite with 1/f-aware noise detection.
- Developed Python drivers and operator GUIs for sample positioning, a Thorlabs Kinesis XYZ piezo stage and a Copley dual-axis stepper drive over RS-232, plus a PyTorch model that filters experimental noise.

Adversarially Robust Federated TinyML for IoT Intrusion Detection — University Park, PA
Honors Research Studies — Jan 2026 – Present

- Built a federated learning pipeline for privacy-preserving intrusion detection on resource-constrained IoT devices, reaching **99% accuracy** with Random Forest and federated models and **96%** with a deep neural network.
- Measure robustness under adversarial evasion attacks, evaluate defenses, and apply pruning and quantization for TinyML deployment on microcontrollers.

PROJECTS

Go Engine from Scratch, AlphaGo-Style MCTS — *Python, PyTorch, MCTS*

- Trained policy and value networks on 10,000 professional games and 500,000 positions, then improved through MCTS self-play to roughly amateur 2-dan strength.
- Diagnosed value-head collapse in self-play and recovered training with resignation logic and expert data mixing.

Catan Reinforcement Learning Agent — *Python, PyTorch, PPO*

- Trained a PPO self-play agent on the Catanatron engine using Penn State CSE GPU servers, reaching a **70% win rate** against the weighted-random baseline.
- Added a relational board encoder, doubled throughput with torch.compile, and shipped a playable browser game.

Research Paper RAG Assistant — *FastAPI, React, ChromaDB, Claude API*

- Built a retrieval-augmented QA service over a paper corpus with vector search and a React UI.

Firefly Platform, AR Autism Education — *LLM decision engine, deployed MVP*

- Shipped an MVP AR and ABA autism intervention platform with an LLM clinical decision engine.